

Intelligent Customer Churn Prediction Using Data Analytics and Classification Models

Project Review-1

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Problem Statement

Customer churn is a major challenge for organizations, as losing existing customers can negatively affect customer relationships and business performance. Machine learning models can be used to predict whether a customer is likely to churn by analyzing demographic, service, usage, contract, and billing-related information. However, customer datasets often contain a large number of variables and complex patterns, making it difficult to identify the key factors associated with customer churn through traditional analysis alone. This creates a need for a systematic data analytics approach that can explore customer behavior, identify important churn-related factors, and accurately classify customers based on their likelihood of leaving. Therefore, there is a need for a comprehensive comparative evaluation of multiple classification models using common customer data and performance metrics to identify effective approaches for customer churn prediction and support data-driven customer retention decisions.

Literature Survey

S.No	Title	Source	Contributor
1.	"Hybrid ensemble learning approaches to customer churn prediction" (2021)	Kybernetes	Proposed hybrid bagging/boosting ensemble classifiers and reported improved churn-prediction performance compared with simpler base classifiers.
2.	"How training on multiple time slices improves performance in churn prediction" (2021) - Gattermann-Itschert & Thonemann	European Journal of Operational Research	Showed that training with multiple historical time slices can improve predictive performance and generalizability in churn prediction.
3.	"An effective strategy for churn prediction and customer profiling" (2022)	Data & Knowledge Engineering	Compared 8 supervised learning methods and 7 sampling approaches across 13 churn datasets; proposed an ensemble-based prediction and customer-segmentation strategy.
4.	"Performance Evaluation of Various Classification Techniques for Customer Churn Prediction in E-commerce" (2022)	Microprocessors and Microsystems	Compared neural networks, SVM, Naive Bayes, Random Forest and deep learning, with PCA and Neighborhood Component Analysis for feature selection.
5.	"Customer churning analysis using machine learning algorithms" (2023)	International Journal of Intelligent Networks	Compared stochastic gradient boosting, Random Forest, KNN and Logistic Regression for early customer-churn prediction.
6.	"Customer retention and churn prediction in the telecommunication industry" (2023) - Saleh & Saha	Discover Applied Sciences	Evaluated Random Forest, AdaBoost, Logistic Regression, XGBoost and Decision Tree across multiple datasets and showed that important churn factors vary across customer populations.

	Title	Source	Contributor
7.	"Research on customer churn prediction and model interpretability analysis" (2023)	PLOS ONE	Developed a GA-XGBoost churn model and included interpretability analysis to identify influential factors behind churn predictions.
8.	"Enhancing customer retention in telecom industry with machine learning driven churn prediction" (2024)	Scientific Reports	Applied machine-learning and ensemble approaches for telecom churn prediction, emphasizing improved customer-retention support.
9.	"Explaining customer churn prediction in telecom industry using tabular machine learning models" (2024)	Machine Learning with Applications	Compared tabular classification models and used SHAP-based global and local explanations to interpret churn predictions.
10.	"Enhancing customer retention with machine learning: a comparative analysis of ensemble models for accurate churn prediction" (2025)	International Journal of Information Management Data Insights	Compared KNN, Logistic Regression, XGBoost and Random Forest using classification metrics, confusion matrices and ROC analysis.
11.	"A data-driven approach with explainable artificial intelligence for customer churn prediction" (2025)	Results in Engineering	Introduced an XAI-based ensemble approach combining XGBoost, CatBoost and LightGBM with feature selection, imbalance handling, SHAP and LIME.
12.	"Forecasting customer churn using machine learning: a comparative modeling approach" (2026)	Neural Computing and Applications	Compared Logistic Regression, Random Forest, SVM, XGBoost, LightGBM, CatBoost, AdaBoost and KNN using Accuracy, Precision, Recall, F1-score and ROC-AUC.

Research Gaps

- ❖ **Limited Integration of End-to-End Churn Analysis** – Most studies focus mainly on prediction performance, with limited integration of EDA, preprocessing, feature engineering, classification, risk scoring, and visualization in one framework.
- ❖ **Inconsistent Handling of Class Imbalance** – Different studies use different techniques such as SMOTE, ADASYN, and ensemble methods to address imbalanced churn data. A consistent comparison of these approaches across multiple models is still needed.
- ❖ **Lack of Comprehensive Comparative Model Evaluation** – Limited studies compare **Logistic Regression, Decision Tree, Random Forest, XGBoost, and CatBoost** using the same dataset, preprocessing steps, and evaluation metrics.
- ❖ **Limited Customer-Level Risk Categorization** – Most studies focus on predicting whether a customer will churn or not. There is a need to convert churn probabilities into **Low, Medium, and High-risk categories** for easier customer prioritization.
- ❖ **Limited Integration of Prediction, Interpretability and Decision Support** – Although SHAP and LIME are increasingly used, prediction, explainability, risk prioritization, and visualization are not always combined in one system. A unified framework can provide more practical churn insights.

Objectives

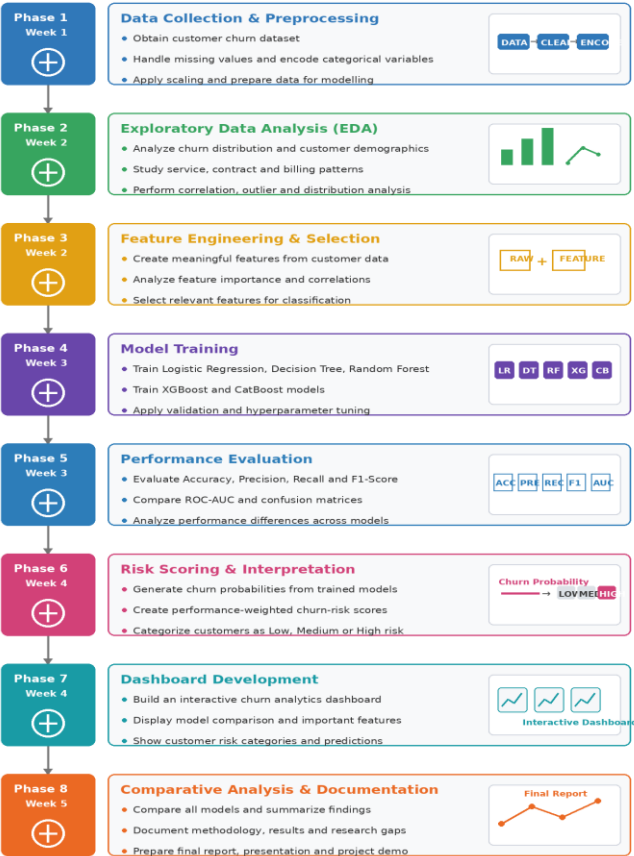
1. To evaluate and compare the predictive performance of five machine learning models (**Logistic Regression, Decision Tree, Random Forest, XGBoost, and CatBoost**) for customer churn prediction.
2. To perform **exploratory data analysis and feature engineering** to identify important customer, service, usage, contract, and billing factors associated with churn.
3. To identify the most influential customer-related factors contributing to **churn risk** using feature importance and model-based analysis.
4. To conduct a comparative analysis of the models using **Accuracy, Precision, Recall, F1-Score, and ROC-AUC**, and categorize customers into **Low, Medium, and High churn-risk levels**.

Innovation & Novelty

S.No	Innovation	Description
1.	Unified Churn Analytics Pipeline	Integrating data quality analysis, EDA, preprocessing, feature engineering, feature selection, classification, risk scoring, and visualization into a single churn prediction pipeline.
2.	Comprehensive Comparative ML Framework	Systematic side-by-side comparison of Logistic Regression, Decision Tree, Random Forest, XGBoost, and CatBoost using the same dataset, preprocessing pipeline, and evaluation criteria.
3.	Performance-Weighted Churn Risk Scoring	Combining model prediction probabilities with model performance to generate a unified churn-risk score and categorize customers into Low, Medium, and High-risk groups.
4.	Interactive Churn Decision-Support Dashboard	Developing an interactive dashboard that presents model performance, important churn factors, customer risk categories, and prediction results to support data-driven customer retention analysis.

Project Workflow & Implementation Plan

Intelligent Customer Churn Prediction Using Data Analytics and Classification Models



Project Plan

Phase 1: Data Collection & Preprocessing

Collect customer churn dataset

Handle missing values and encode categorical variables

Normalize features and apply SMOTE

Phase 2: Exploratory Data Analysis (EDA)

Perform 12+ EDA techniques

Analyze churn distribution, customer patterns and correlations

Identify outliers and important relationships

Phase 3: Feature Engineering & Selection

Create new features from existing customer data

Select important features using correlation and feature importance

Phase 4: Model Training

Train **Logistic Regression, Decision Tree, Random Forest, XGBoost and CatBoost**

Apply 5-fold cross-validation and hyperparameter tuning

Phase 5: Performance Evaluation

Evaluate using **Accuracy, Precision, Recall, F1-Score and ROC-AUC**

Apply Borda Count for model ranking

Phase 6: Churn Risk Scoring

Generate churn probabilities from the trained models

Categorize customers into **Low, Medium and High risk**

Identify important churn factors

Phase 7: Comparative Analysis & Dashboard

Compare model performance and churn predictions

Develop an interactive dashboard for customer risk analysis

Document the findings and results in the final report